

1 Solution — GIAM 1.2

Problem 9

1.1 Problem

A famous conjecture thought to be true (but for which no proof is known) is the Twin Prime conjecture: a pair of primes is *twin* if they differ by 2 (e.g. 11 and 13, or 431 and 433), and the conjecture asserts there are infinitely many. Try to come up with an argument as to why 3, 5, 7 are the only prime *triplet*.

1.2 Setup — What a “Prime Triplet” Means Here

By analogy with twin primes, a **prime triplet** here means three primes that each differ by 2 — i.e. three primes of the form

$$n, \quad n + 2, \quad n + 4,$$

three consecutive odd numbers all prime. The claim is that the *only* such triplet is 3, 5, 7. Unlike the twin prime conjecture, this one has a short, complete proof.

1.3 The Key Fact — One of $n, n + 2, n + 4$ Is Always Divisible by 3

Look at the three numbers modulo 3. Since $4 \equiv 1 \pmod{3}$,

$$n \equiv n, \quad n + 2 \equiv n + 2, \quad n + 4 \equiv n + 1 \pmod{3}.$$

So modulo 3 the three numbers are $n, n + 1, n + 2$ in some order — *three consecutive residues*, which run through **all** of 0, 1, 2. Therefore exactly one of $n, n + 2, n + 4$ is $\equiv 0 \pmod{3}$:

Among any three numbers $n, n + 2, n + 4$, exactly one is divisible by 3.

1.4 The Argument

Suppose $n, n+2, n+4$ are *all prime*. By the key fact, one of them is divisible by **3**. But a prime divisible by **3** can only be **3** itself (any other multiple of **3** is composite). So one of the three numbers must equal **3**. There are three cases:

- $n = 3$: the triplet is **3, 5, 7** — and indeed all three are prime. ✓
- $n + 2 = 3$, i.e. $n = 1$: the triplet is **1, 3, 5** — but **1** is not prime. ✗
- $n + 4 = 3$, i.e. $n = -1$: not a positive integer. ✗

Only the first case yields three primes. Therefore

3, 5, 7 is the unique prime triplet. ■

1.5 Seeing It in a Table

Every triplet $(n, n+2, n+4)$ contains a multiple of 3. Only when that multiple is 3 itself can all three be prime.

n, n+2, n+4			multiple of 3 (highlighted)
3	5	7	3 is the only prime multiple of 3 → ✓ valid: 3, 5, 7
5	7	9	9 = 3 · 3 is composite → ✗
7	9	11	9 = 3 · 3 is composite → ✗
9	11	13	9 = 3 · 3 is composite → ✗
11	13	15	15 = 3 · 5 is composite → ✗
17	19	21	21 = 3 · 7 is composite → ✗

■ multiple of 3 that IS prime (only 3)
 ■ multiple of 3 that is composite

Figure 1.1: Every triplet $(n, n+2, n+4)$ has exactly one member divisible by 3 (highlighted). For 3, 5, 7 that member is 3 itself — prime — so the triplet survives. For every other triplet the multiple of 3 is at least 9, hence composite, killing the triplet.

The highlighted cell in each row is the member divisible by **3**. In the very first row it is **3**, the lone prime multiple of **3**, so **3, 5, 7** all pass. In *every other* row the multiple of **3** is **9, 15, 21, ...** — composite — so the triplet fails. Notice the multiple of **3** moves around (first, middle, or last position) depending on $n \bmod 3$, but it is *always present*.

1.6 Remark — Why Triplets Are Settled but Twins Are Not

It is striking that this triplet question is completely *solved* by a one-line modular argument, while the closely related Twin Prime conjecture (about *pairs* $p, p + 2$) remains famously open. The difference is the mod-3 obstruction:

- For a **pair** $p, p + 2$: modulo 3 these are two residues, which need *not* include 0. (E.g. $p \equiv 2$ gives $p \equiv 2, p + 2 \equiv 1$ — neither divisible by 3.) So there is no divisibility obstruction, and pairs of primes keep appearing as far as anyone has checked — hence the conjecture that they never stop.
- For a **triplet** $p, p + 2, p + 4$: modulo 3 these cover all three residues, *forcing* one to be divisible by 3. This obstruction caps the triplets at the single exception 3, 5, 7.

So the triplet problem is easy precisely *because* three terms cannot dodge the modulus 3, whereas two terms can. The twin prime problem is hard precisely because no such obstruction exists to exploit.

1.7 Remark — How Number Theorists Re-Define “Prime Triplet”

Because $(p, p + 2, p + 4)$ is doomed by the mod-3 wall, professional number theory defines a **prime triplet** differently — as three primes in the tightest possible cluster that *avoids* a complete residue system mod 3:

$$(p, p + 2, p + 6) \quad \text{or} \quad (p, p + 4, p + 6).$$

These patterns hit only *two* residues mod 3, so they have no forced multiple of 3 and can occur infinitely often (conjecturally). Examples: (5, 7, 11), (7, 11, 13), (11, 13, 17), (13, 17, 19). It is conjectured — but, like the twin prime conjecture, *not proved* — that there are infinitely many of these. The pattern $(p, p + 2, p + 4)$ is deliberately *excluded* from the definition precisely because of the result we just proved.

1.8 Remark — The Same Idea Kills Other “Naïve” Patterns

The covering-all-residues trick is a general tool. Any cluster of integers whose differences make them cover every residue class modulo some prime q can contain at most one prime larger than q . For instance:

- $n, n+2, n+4, n+6$ — modulo 3 this still covers $\{0, 1, 2\}$ (the first three already do), so again a multiple of 3 appears; four primes in a row at gap 2 is impossible beyond the start.
- $n, n+1$ — modulo 2 one is always even, so the only “consecutive-integer prime pair” is $2, 3$.

The same one-line reasoning (“these residues cover everything mod q , so q divides one of them; for all to be prime that one must be q ”) settles each case. It is the cleanest weapon in elementary prime patterns — and a reminder that some prime questions are easy for a *structural* reason, even when their cousins (like twin primes) are deep.

1.9 Remark — $3, 5, 7$ Is Also the Only Run of Three Consecutive Odd Primes

Restating the result in plain terms: $3, 5, 7$ is the **only set of three consecutive odd numbers that are all prime**. Every later block of three consecutive odd numbers $\{n, n+2, n+4\}$ (with $n \geq 5$) is interrupted by a multiple of 3 that exceeds 3 and is therefore composite. This makes $3, 5, 7$ a genuinely special object in the integers — the unique “prime triplet” — sitting right at the start of the number line where 3 itself can play the role of the unavoidable multiple of 3 .

1.10 Remark — Longer Prime Progressions Need Bigger Gaps

The result does *not* say prime arithmetic progressions of length 3 are impossible — only that ones with common difference 2 are. With a larger common difference the mod- 3 obstruction can be avoided: for example

$$5, 11, 17 \quad (\text{difference } 6), \quad 7, 37, 67 \quad (\text{difference } 30)$$

are length- 3 arithmetic progressions of primes. In fact the **Green–Tao theorem** (2004) proves there are arbitrarily long arithmetic progressions consisting entirely of primes. But their common differences must be divisible by all small primes (for length ≥ 3 , by 2 and 3 ; for the AP to avoid a forced multiple of 3 the difference must be a multiple of 3 , etc.) — which is exactly why the gap- 2 case collapses to the single starter $3, 5, 7$.